

Scalability & Future Plan

AI-Assisted Smart Budget Planner

Current System Limitations

S.No	Limitation	Priority to Address	Impact
1	Catalog prices are illustrative and not live-verified.	High	Users must confirm current merchant prices.
2	SQLite is intended for the current/local scope.	Medium	Larger multi-user deployments may require a persistent production database.
3	No dedicated load-test measurements are documented.	Medium	Capacity under high concurrent load is not established.
4	Gemini is optional and requires an API key for AI commentary.	Low	Without a key, AI commentary is unavailable but planning still works.

Scalability Plan

Aspect	Current State	Proposed Upgrade / Solution
User Load	Local FastAPI + SQLite	Deploy behind a production ASGI setup and add rate limiting/load testing.
Data Storage	SQLite	Move to a persistent production database with backups.
Performance	Deterministic packing; no load benchmark documented	Add load/stress testing, caching where useful, and monitor API latency.
Security	scrypt, HTTP-only cookies, JWT	HTTPS, secure cookies, rate limiting, secret management and monitoring.

Future Roadmap

Phase	Planned Feature / Enhancement	Expected Impact
Phase 2	Live price/catalog integration through approved APIs	More current estimates
Phase 3	Production database and deployment	Higher reliability and multi-user scale
Phase 4	Advanced personalization / richer Gemini assistance	More contextual recommendations